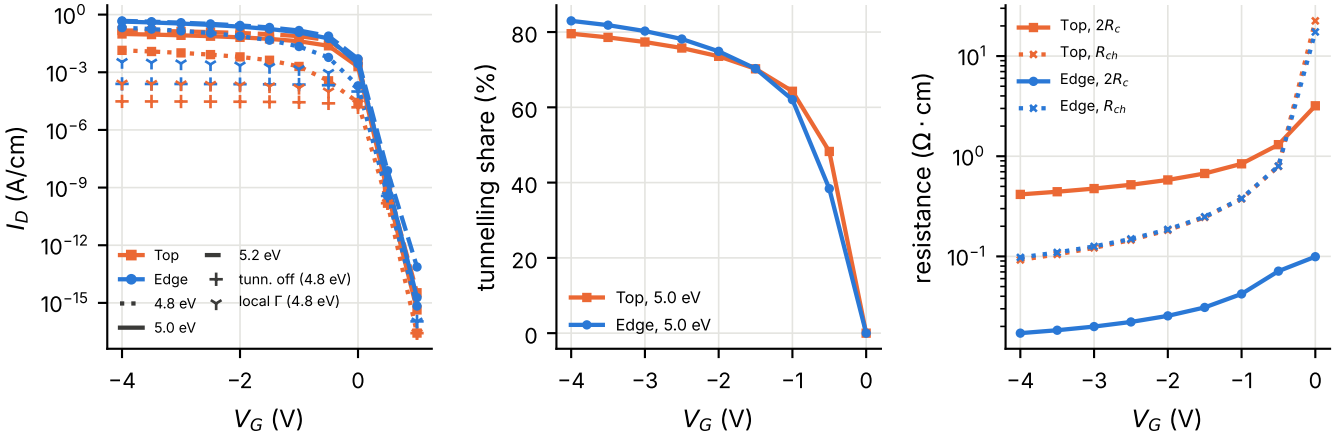
(a) Transfer, V_D=0.05 V
I_D (A/cm)
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
Top
Edge
4.8 eV
5.0 eV
5.2 eV
tunn. off (4.8 eV)
local Γ (4.8 eV)
−4
−2
0
V_G (V)
(b) Injection by tunnelling
tunnelling share (%)
80
60
40
20
0
Top, 5.0 eV
Edge, 5.0 eV
−4
−3
−2
−1
0
V_G (V)
(c) Contact vs channel, 5.0 eV
resistance (Ω · cm)
10^{1}
10^{0}
10^{-1}
Top, $2R_c$
Top, R_{ch}
Edge, $2R_c$
Edge, R_{ch}
−4
−3
−2
−1
0
V_G (V)